Project Documentation – Quiz Game

1. PROJECT OVERVIEW

1.1 Project Identification

Project Name: Quiz Game

Project Type: Software

1.2 Project Team

Project Lead: Sahil Abdulalim

Team Members: N/A

Supervisor/Advisor: Shaukatali Hussein

1.3 Project Summary

Description of Project:

This project is a web-based trivia application that challenges users with [Insert Topic, e.g., General Knowledge/Coding] questions. The primary goal was to create a "UI-first" experience—ensuring that transitions, feedback states, and layouts are intuitive and visually engaging.

Problem Statement:

The current landscape of online quiz platforms is often hindered by cluttered interfaces, poor mobile responsiveness, and a lack of immediate, intuitive feedback, all of which create unnecessary cognitive load for the user

Objectives:

Main Objective: The primary goal of this project is to bridge the gap between educational content and high-end digital interaction by developing a **UI-centric quiz platform**.

Specific Objectives:

1. To design a minimalist, "distraction-free" interface that keeps the user focused on one question at a time.

2. To ensure 100% functional and visual parity across mobile, tablet, and desktop breakpoints.

2. TECHNICAL SPECIFICATIONS

2.1 Hardware Components

N/A

2.2 Software Components

Item Name	Quantity / Unit
HTML	1
CSS	1
JavaScript	1

2.3 Tools & Equipment Used

Fabrication Equipment:
N/A

Software Tools:

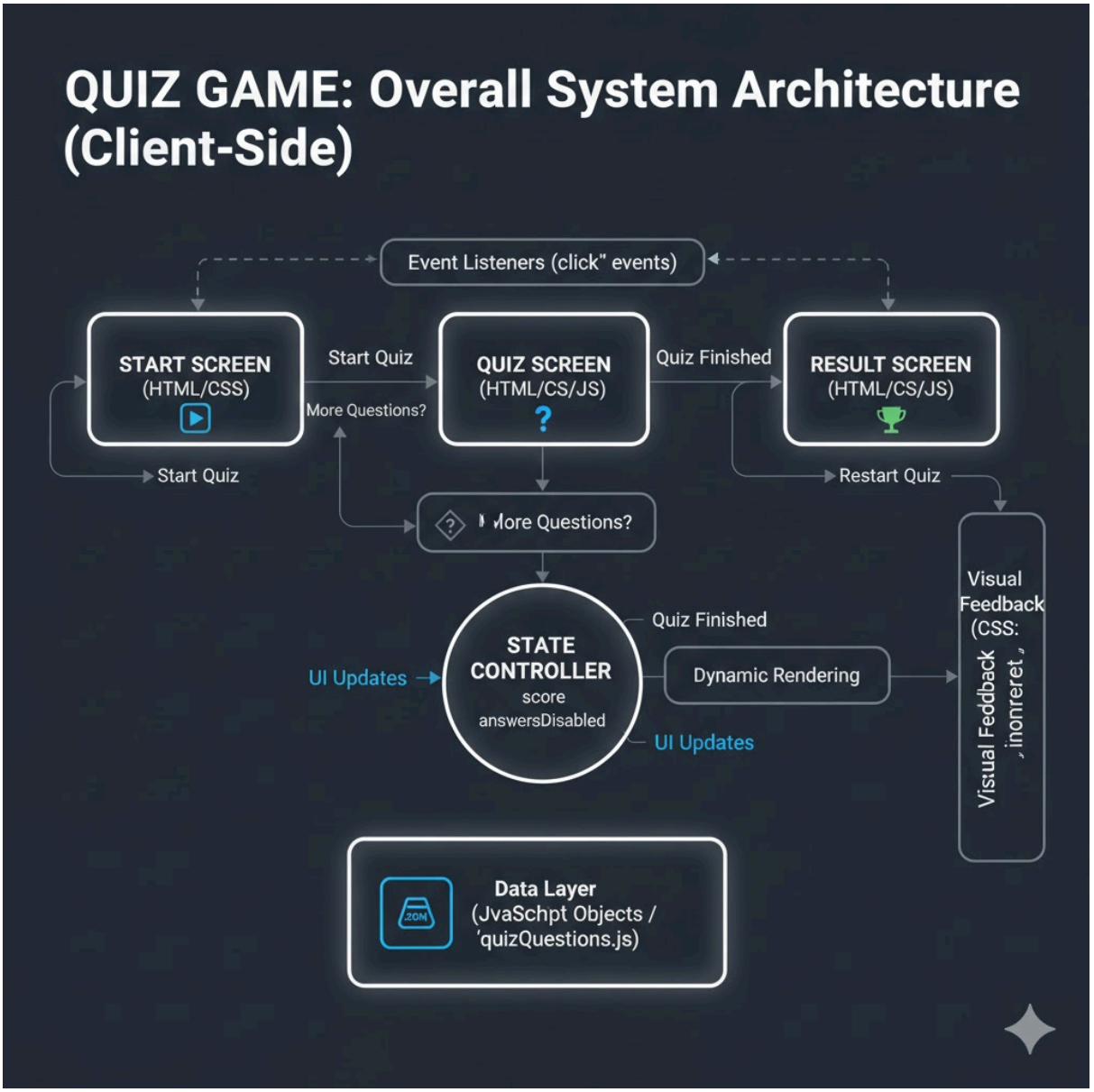
Tool	Use
HTML	Main programming language for developing the game
CSS	Styling the Quiz Game
JavaScript	Responsiveness of the Quiz Game
VS Code	Development and debugging

GitHub	Version control

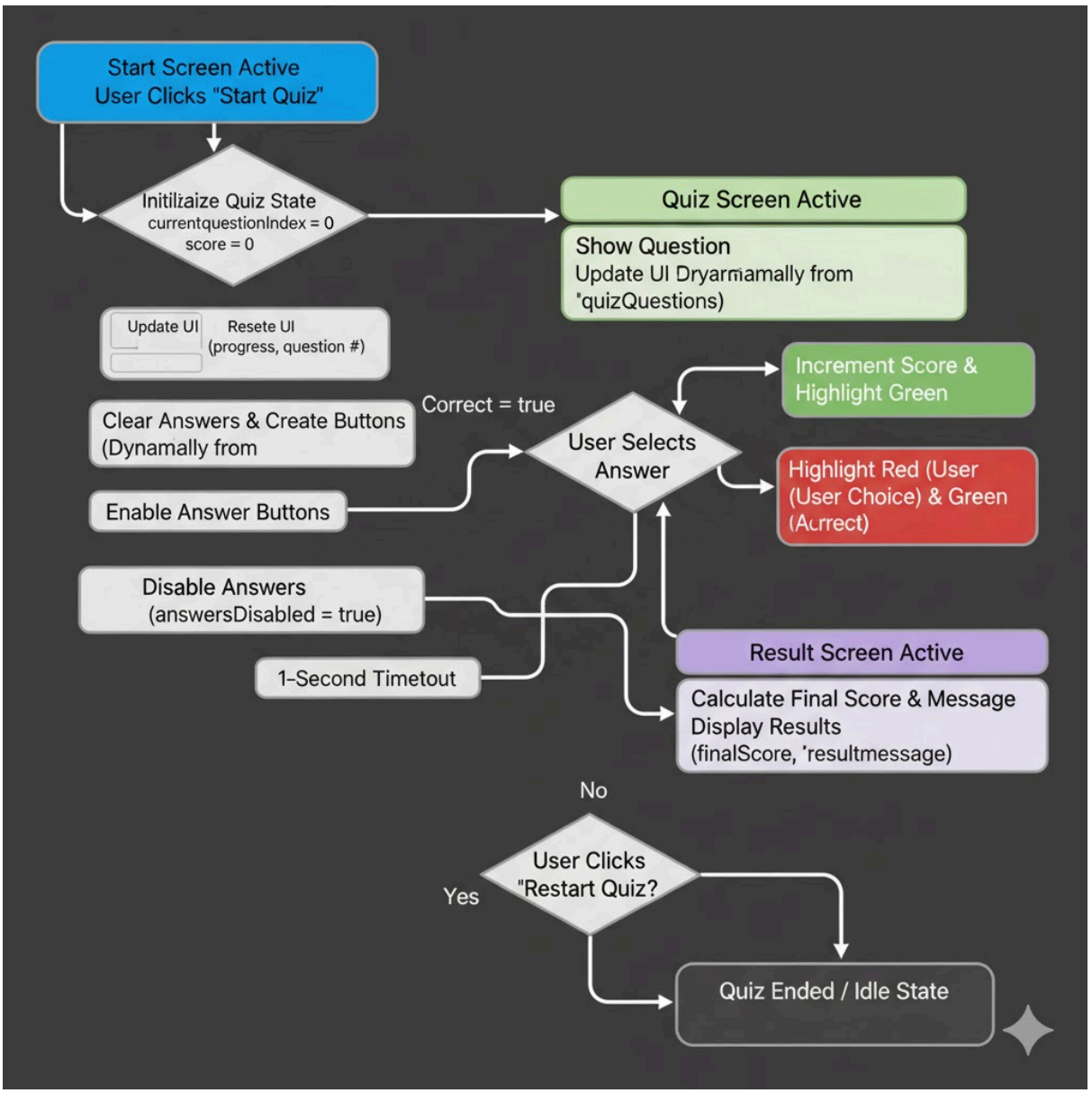
3. DESIGN DOCUMENTATION

3.1 System Architecture

Overall System Diagram:



Flowchart/Process Flow:



3.2 Hardware Design

N/A

3.3 Software Design

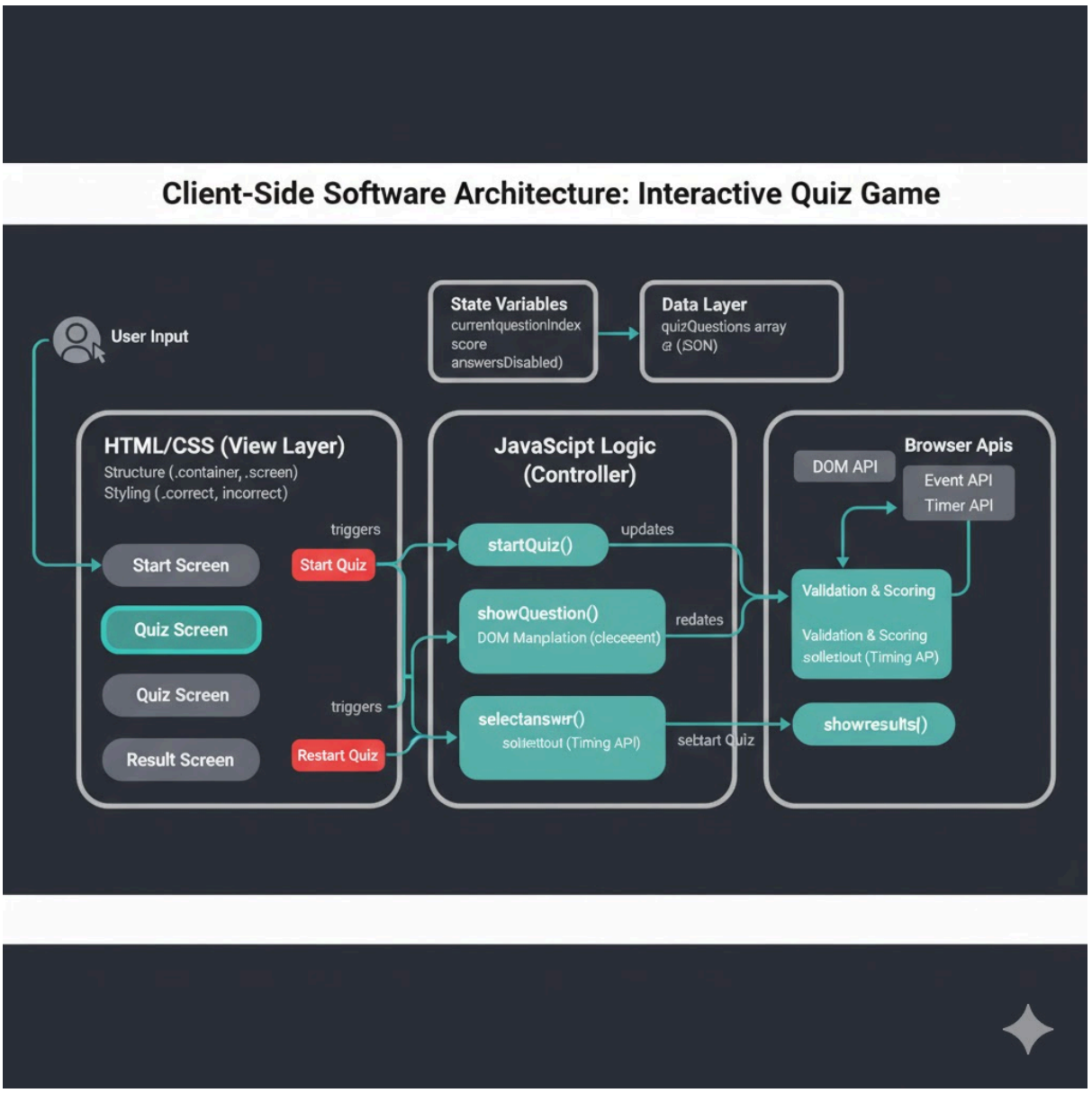
Programming Language(s): HTML, CSS and JavaScript

Code Repository: https://github.com/TOIO-STUDENTS-PROJECTS/Quiz-Game

API Documentation: N/A

Database Schema: N/A (Game does not use a database)

Software Architecture Diagram:



5. BILL OF MATERIALS (BOM)

Item	Description	Quantity	Unit Cost	Total Cost
Laptop	For Development	1	Existing	0
HTML, CSS + JS	Open-source	1	Free	0

Total Project Cost: 0 Tzs (Software-only project)

6. PROJECT EVALUATION

6.1 Success Criteria

Objectives Met:

Objective 1:Maintained a processing latency of less than **50ms** to prevent "input lag" during selection.

Objective 2: Users were able to navigate the quiz without instructions, proving the "UI-First" objective was met.

Performance vs. Expectations:

Instant feedback on every click. This worked as it was expected.

6.2 Lessons Learned

What Worked Well:

- The app uses a central State Controller to track progress and performance
- Using CSS variables (`:root`) allows you to manage the entire visual theme from a single location.

What Could Be Improved:

- Implement the **Web Storage API (localStorage)**. This would allow you to save the user's high scores across browser refreshes.
- Move the `quizQuestions` array into a separate `data.json` file. Use the **Fetch API** to load this data asynchronously, which decouples your content from your logic.

Skills Acquired:

- Structuring an application with clear, logical container hierarchies.
- Mastery of CSS variables (`:root`), Flexbox/Grid for responsive layouts, and state-based styling using class toggling.

- Using `@media` queries to ensure a consistent user experience across mobile and desktop devices.

Recommendations for Future Projects:

- Instead of a hardcoded array, fetch questions from a live server.
- "High-Score" Leaderboard (Persistence)

- Personal Dashboard

6.3 Future Work

Potential Enhancements:

1. Add difficulty levels
2. Save high scores using a database or local storage.

Scalability Considerations:

- Can be adapted for experiential marketing.

Next Steps:

- Add multiplayer and ranking mode for competition.

7. PRESENTATION & OUTREACH

7.1 Documentation for Showcase

Project Photos: Not yet
Demo Videos: Not yet
Presentation Slides: Not yet